

虛擬教室期末報告

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專題摘要

透過unity開發虛擬教室環境，結合大語言模型讓老師與虛擬學生互動，從互動過程中學習應對各種學生(包括過動生、正常生)的突發狀況。系統主要包含兩部分，UDL系統及NDBI系統。

UDL旨在開發**教室設置系統**，目的為訓練老師能夠清楚的理解教室應該要有的合理配置，以及避免應設施擺設不當造成學生受傷的潛在危險。

NDBI旨在開發**實際教學互動**場景，透過解決以及發現虛擬學生的問題，訓練老師對不同學生的教學方式變化和習慣問題生可能引發的問題。

中教大計畫摘要

- **【子計畫一】 NDBI 全作息自然情境教學訓練系統**

訓練教師在「全日作息」中，利用環境配置誘發(由老師發起的誘發事件)幼兒溝通動機，並執行 NDBI 策略與自然增強。

- **【子計畫二】 情緒行為共情與生理狀態評量系統**

監測教師面對幼兒挑戰行為時的生理壓力應激，並分析其對話內容的共情品質。

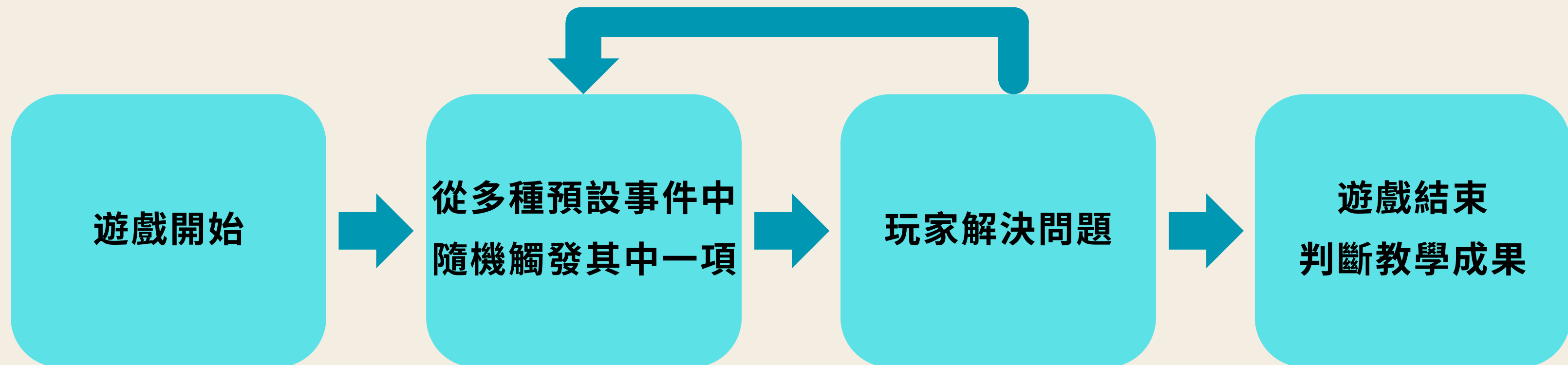
- **【子計畫三】 通用設計學習(Universal Design for Learning, UDL) 物理環境規劃與動線分析**

開發一套具備教育訓練目的的沙盒模擬遊戲 (Serious Game)，讓使用者在拆解任務的過程中，自然內化通用設計學習(UDL)的核心原則

事件觸發版本差異(NDBI介紹)

1. 被動觸發

每隔一段時間自動觸發LLM生成的隨機搗蛋或是特定狀況，老師需要做出對應情況的反應以解決突發事件，例如安撫學生情緒以及指揮學生動作。

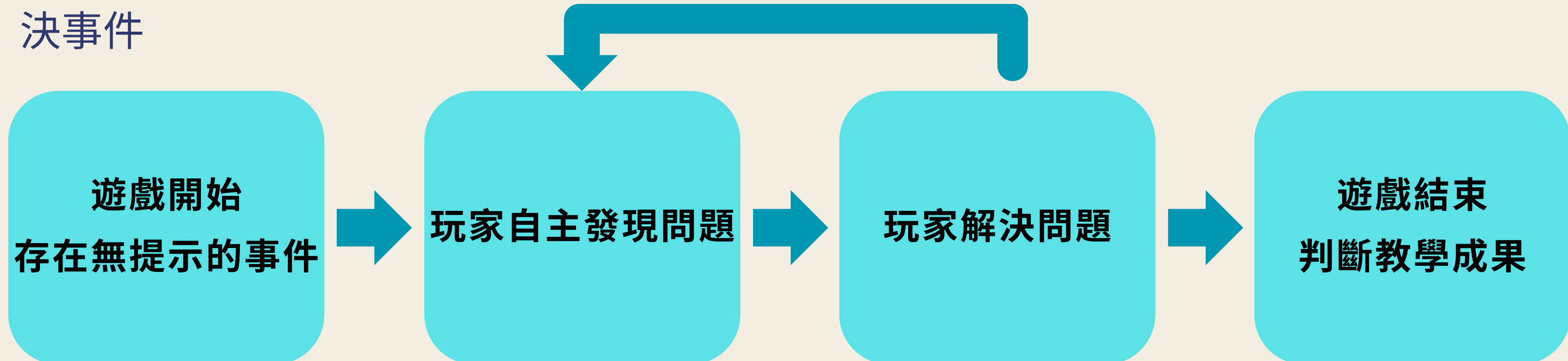


事件觸發版本差異(NDBI介紹)

2. 主動觸發(NDBI)

為了因應現實中部份學生的需求無法妥善表達，老師需要主動發現潛在的教學機會，例如發現學生的異樣或是需求並加以解決。

教室中同時存在多樣待發現的事件，老師須自行察覺異樣並做出適當行為解決事件



成果展示

詳細分工

廖聿翔:

- 擔任組長組織開會、討論及工作分配
- 情緒偵測模組
- UDL系統功能開發

楊敦棋:

- LLM串接(gemini)
- 回覆記憶系統
- LLM對答系統設計(被動事件、主動觸發事件)

陳喬予:

- 與美術人員協作完成場景所需3D物件
- 教室物件設置
- 繪本等可互動物件

張凱翔:

- 連續語音輸入系統
- 語音轉文字
- UI介面
- 遊戲運作腳本(動畫、巡路等)

THANK YOU

MY PROFILE

Ketut Susilo

As a dedicated professional with a passion for visual communication, I bring a unique blend of creativity and strategic thinking to every project. My expertise lies in transforming complex ideas into compelling visual narratives that resonate with audiences and drive engagement.



MOTIVATION動機

October 2022:

Initial Concept & Research: Laying the groundwork for innovative visual strategies.

June 2030:

Implementation & Execution Phase: Bringing the vision to life with precision and creativity.

May 2025:

Strategic Planning & Development: Refining methodologies and developing core frameworks.

August 2032:

Evaluation & Future Scaling: Assessing impact and planning for sustained growth and evolution.

INTRODUCTION

In today's dynamic landscape, visual impact is paramount. This proposal outlines a transformative project designed to leverage cutting-edge visual graphics to solve challenges and create unparalleled opportunities. We aim to set new benchmarks in how information is conveyed and absorbed.

PROJECT TIMELINE

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OUR EXPERT TEAM

Our collective strength lies in diverse expertise and a shared commitment to excellence. Each team member brings a unique skill set, ensuring a holistic and highly effective approach to every facet of the project.



Avery Davis



Estelle Darcy



Pedro Fernandes



Morgan Maxwell

OBJECTIVES

Objective 1

To develop and implement a robust visual graphics framework that enhances user engagement and clarifies complex data, ensuring seamless information delivery across all platforms.



Objective 2

To establish a scalable and adaptable system for visual content creation, enabling rapid response to evolving market needs and maintaining a competitive edge through visual innovation.



IDEA ANALYSIS

Audience

Tailoring visuals for maximum resonance.

Data Visualization Mastery

Transforming raw data into compelling stories.

Interactive Experiences

Engaging users through dynamic visuals.

Brand Consistency

Ensuring visual harmony across all touchpoints.

Accessibility Focus

Designing for inclusivity and broad reach.

Emerging Technologies Integration

Leveraging new tools for greater impact.

Performance Measurement

Tracking engagement and visual effectiveness.

Continuous Improvement Loops

Iterating based on feedback and results.

CONCEPTUAL FRAMEWORK

Brilliant Idea

Leveraging cutting-edge technology for superior visual output and streamlined workflows.

Amazing Idea

Illuminating complex concepts with clarity and simplicity, making them universally understandable.

Marvelous Idea

Fostering collaborative visual development through intuitive tools and real-time feedback.

Original Idea

Pioneering new visual aesthetics and techniques to differentiate and inspire.

Creative Idea

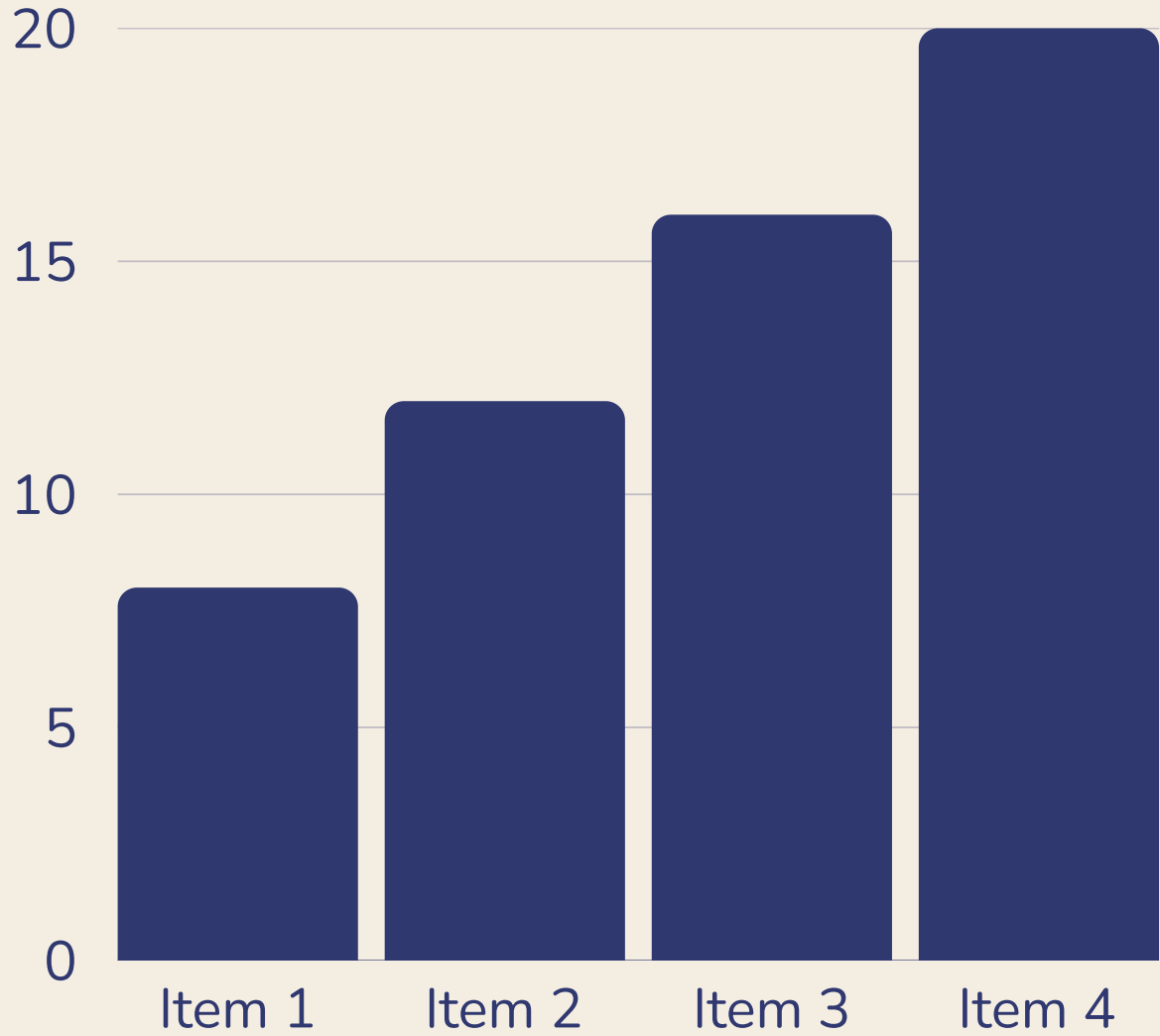
Crafting unique visual narratives that break through the noise and capture attention.

Ingenious Idea

Strategically targeting visual content to specific audience segments for maximum impact.

VISUAL INFOGRAPHIC

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CONCLUSION

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October 2030

THANK YOU

Thanks For Watching

Presented by Ketut Susilo